

ECEN 474/704: Lab 08

Operational Transconductance Amplifier(Designing OTA based on specs)

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Due date: 2025-11-21

I. DISCUSSION OF THE LAB AND RESULTS

Summary of Pre-Layout and Post-Layout Analysis

In the pre-lab analysis, all transistors were assumed to operate in saturation, and the design parameters were driven primarily by the power and slew-rate specifications. The bias current (**I_{bias}**) was initially calculated based on the required slew rate. During pre-layout simulation, the circuit achieved a gain of **29.515 dB**, a unity-gain bandwidth of **6.57 MHz**, power consumption of **486 μ W**, and a slew rate of **8.22 V/ μ s**. After layout extraction, the post-layout results showed a slight degradation due to parasitic effects: gain reduced to **28.37 dB**, UGB decreased to **6.12 MHz**, power dropped to **450 μ W**, and the slew rate reduced to **8.12 V/ μ s**. Phase margin also decreased from **91.8°** (pre-layout) to **83.17°** (post-layout), which is expected given the additional capacitances introduced by layout parasitics. During simulation, it was observed that **increasing I_{bias} improved the slew rate**, as expected from the relation $SR = I_{bias}/C$. However, this also increased the overall power consumption, creating a clear **trade-off between power and slew rate**. Similarly, adjusting the **PMOS current mirror dimensions** (while maintaining the 1:2 current mirror ratio) helped increase the output swing, but shrinking device sizes also influenced current capability and parasitic loading. To meet the design specifications, a **trial-and-error refinement** of transistor dimensions and bias current was performed. By carefully balancing device sizing, current mirror ratios, and I_{bias}, the final design achieved the required power and slew-rate specifications while minimizing post-layout performance loss.

II. SUMMARY OF RESULTS

Summarizing the sizes of the transistors used:

M1	4u	270n
M2	4u	270n
M3	2u	270n
M4	2u	270n
M5	4u	270n
M6	4u	270n
M7	8u	270n
M8	8u	270n
M9	8u	270n
M10	8u	270n
ibias	90uA	

Table 1

Example summary table of pre- and post-layout results

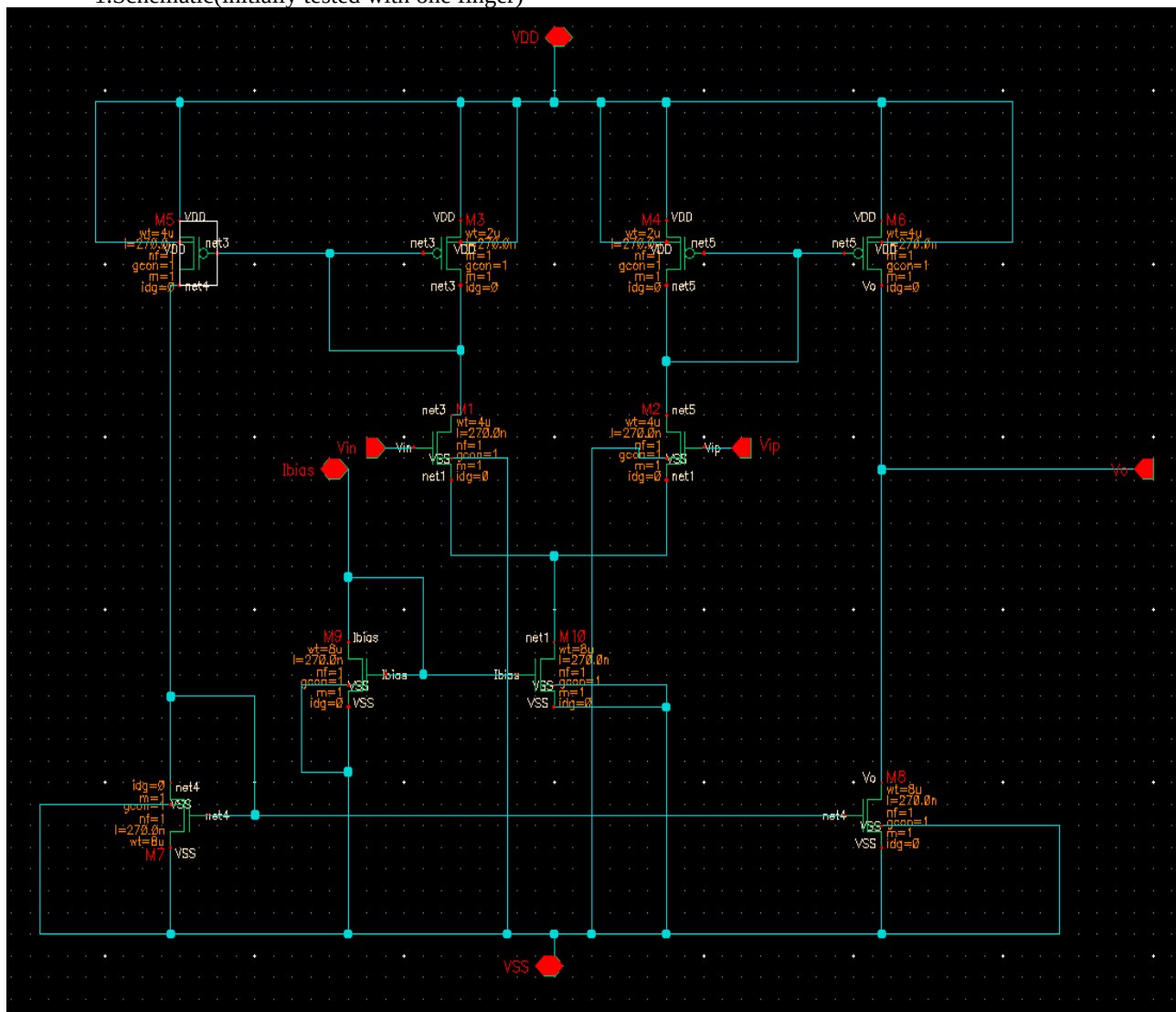
Parameter	Gain	GBW	Power	Open loop Voltage gain	Output swing	Open loop transconductance-GM	Dominant pole frequency	Slew rate	Phase margin
Pre-layout	29.515db	6570.35(10 ^{^(29.515/20)})*233.125kHz	270.2592*1.8=486uW	28.951 db	1.30172	866.395uA/V	233.125kHz	8.22455V/us	91.804deg

Parameter	Gain	GBW	Power	Open loop Voltage gain	Output swing	Open loop transconductance-GM	Dominant pole frequency	Slew rate	Phase margin
Post-layout	28.371db	6119.94	250.2947*1.8=450uW	28.0087db	1.212	703.8uA/V	233.479kHz	8.1169V/us	83.174deg

III. SUB-CIRCUIT AND TEST BENCH SCHEMATICS

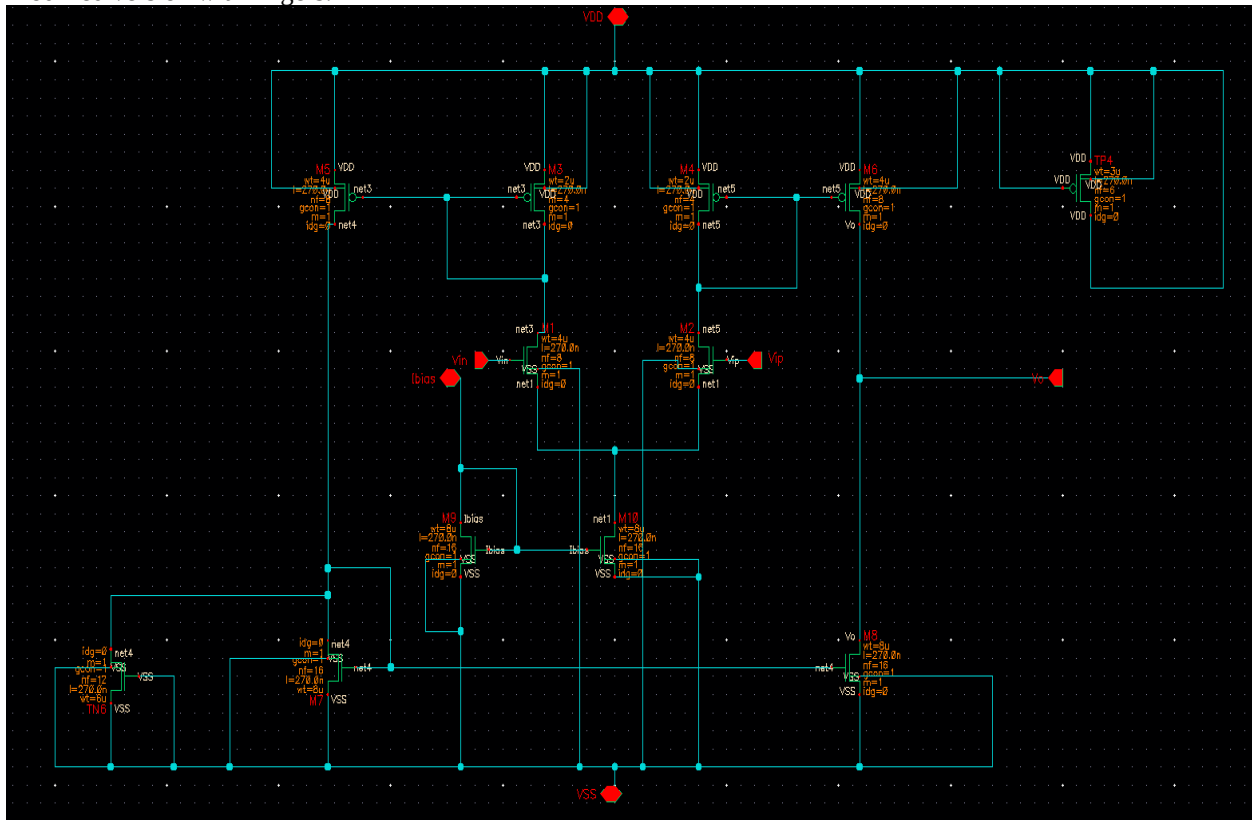
Schematics of the sub-circuit(s) being designed in the lab should be included here. In addition, include images of the test bench schematics. Important signals in the test bench schematic should be labeled to provide clear correspondence between the signals in the schematics and the signals that are plotted in the simulation results shown later in the report.

1.Schematic(initially tested with one finger)

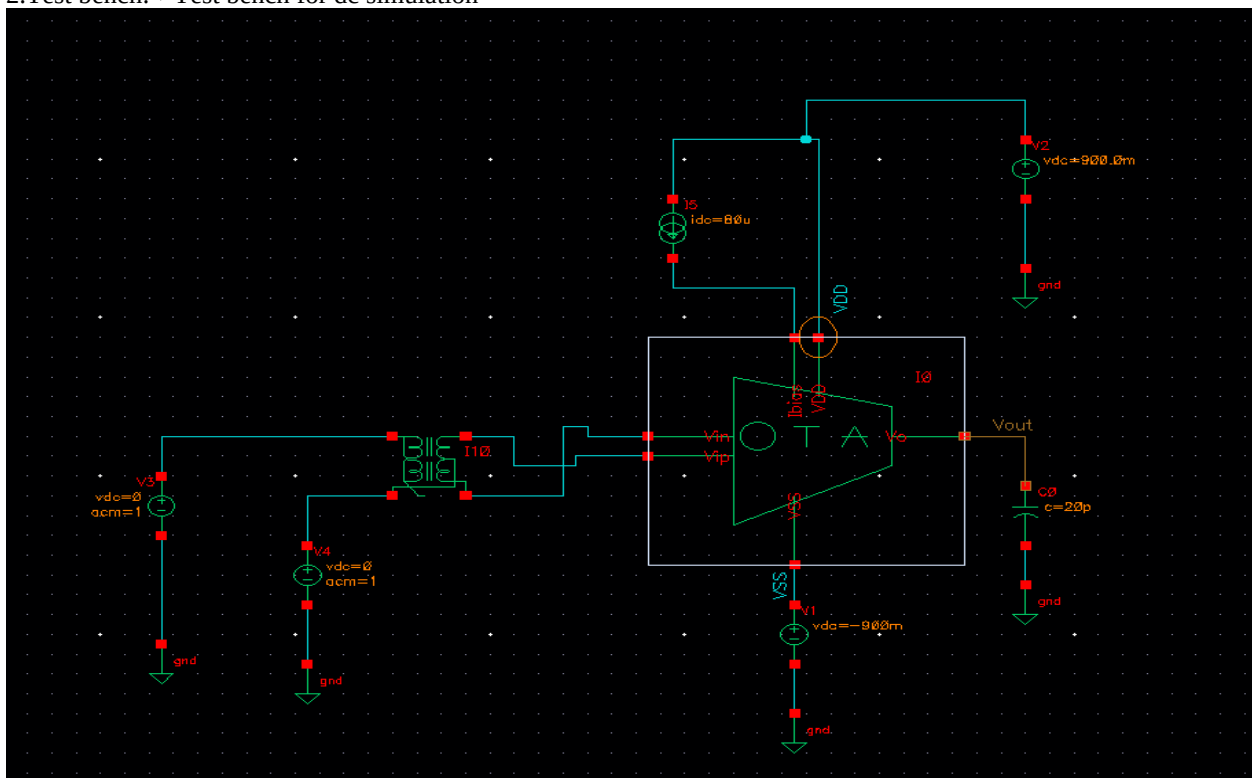


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Modified version with fingers:

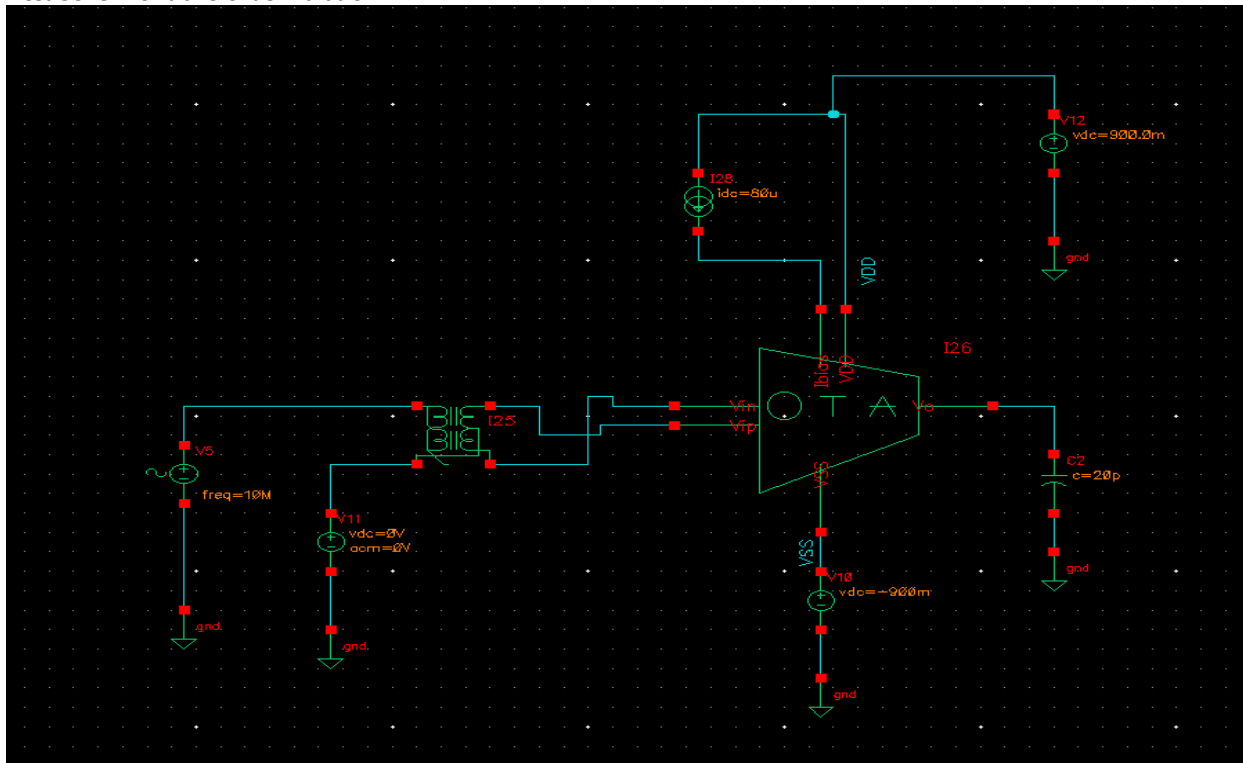


2. Test bench:->Test bench for dc simulation

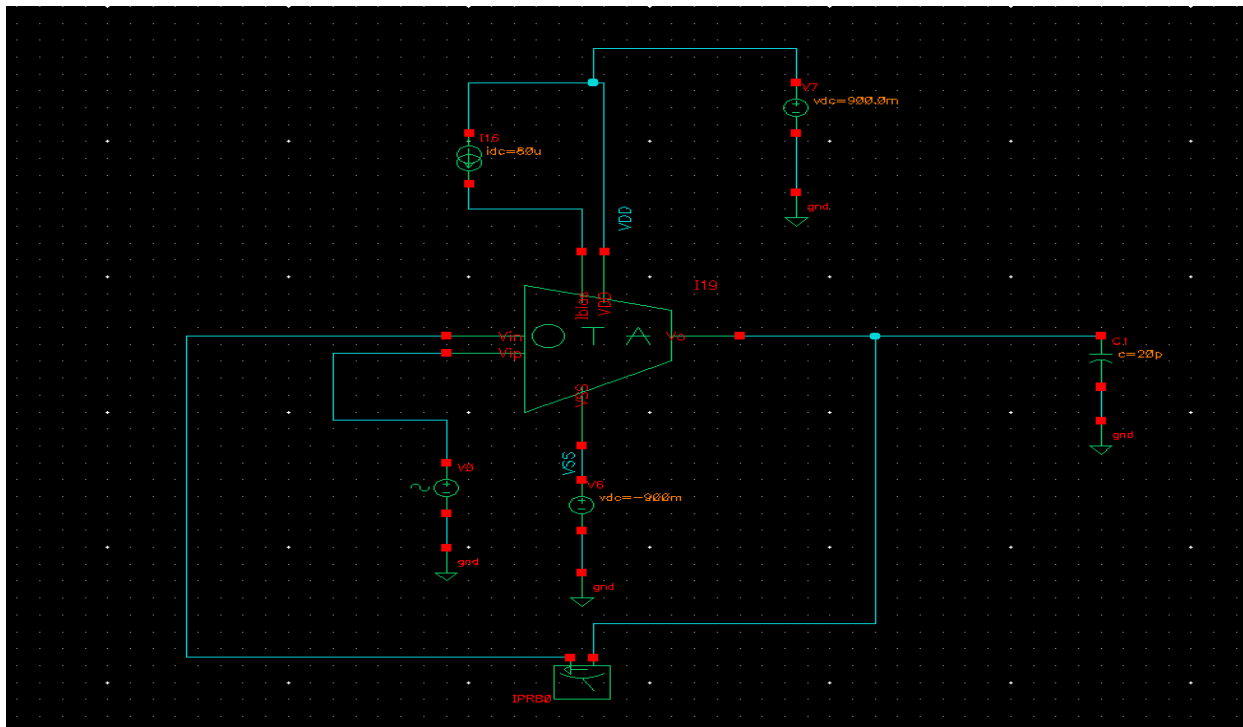


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Test bench for transient simulation

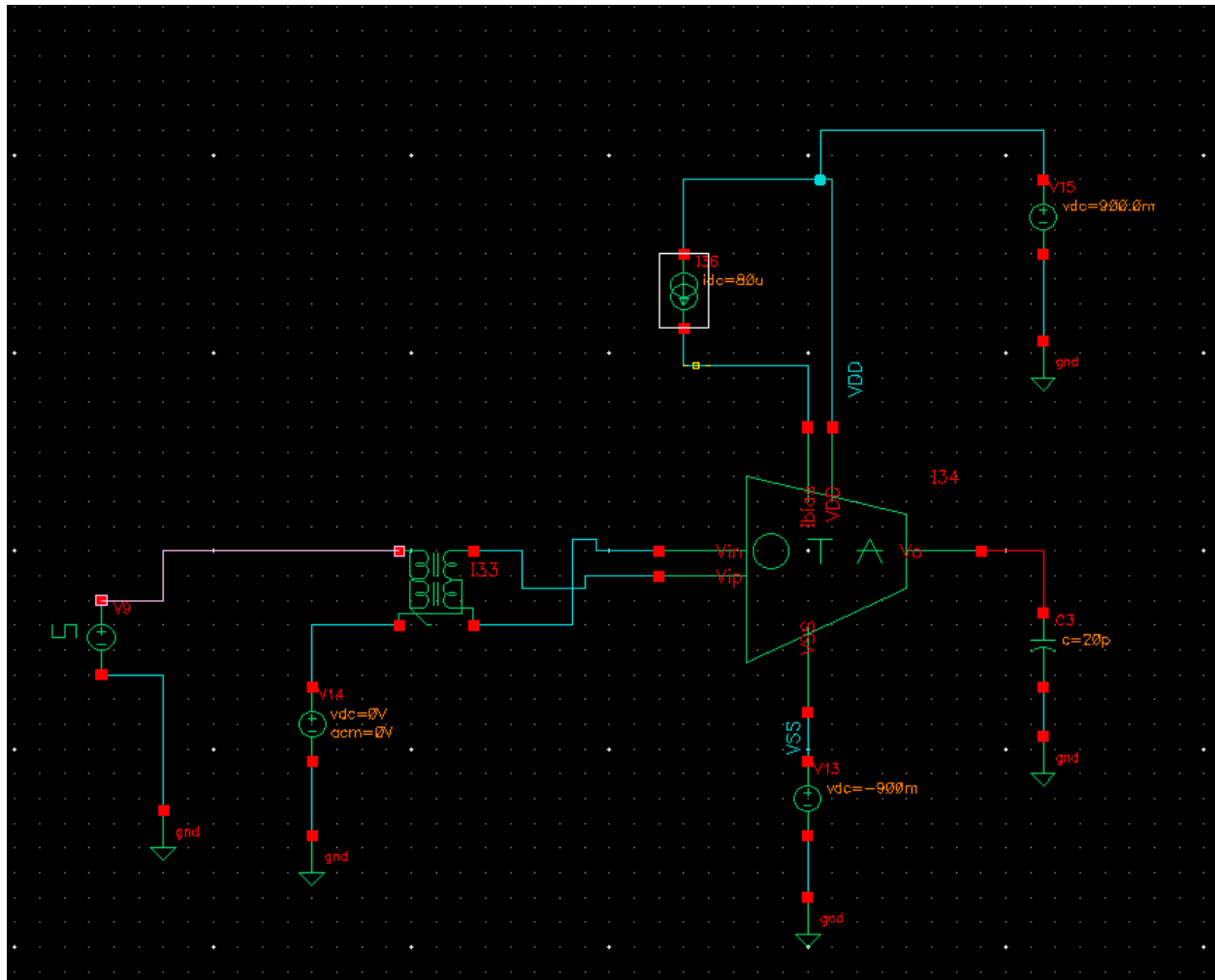


Test bench for closed loop gain



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Test bench for transient simulation:



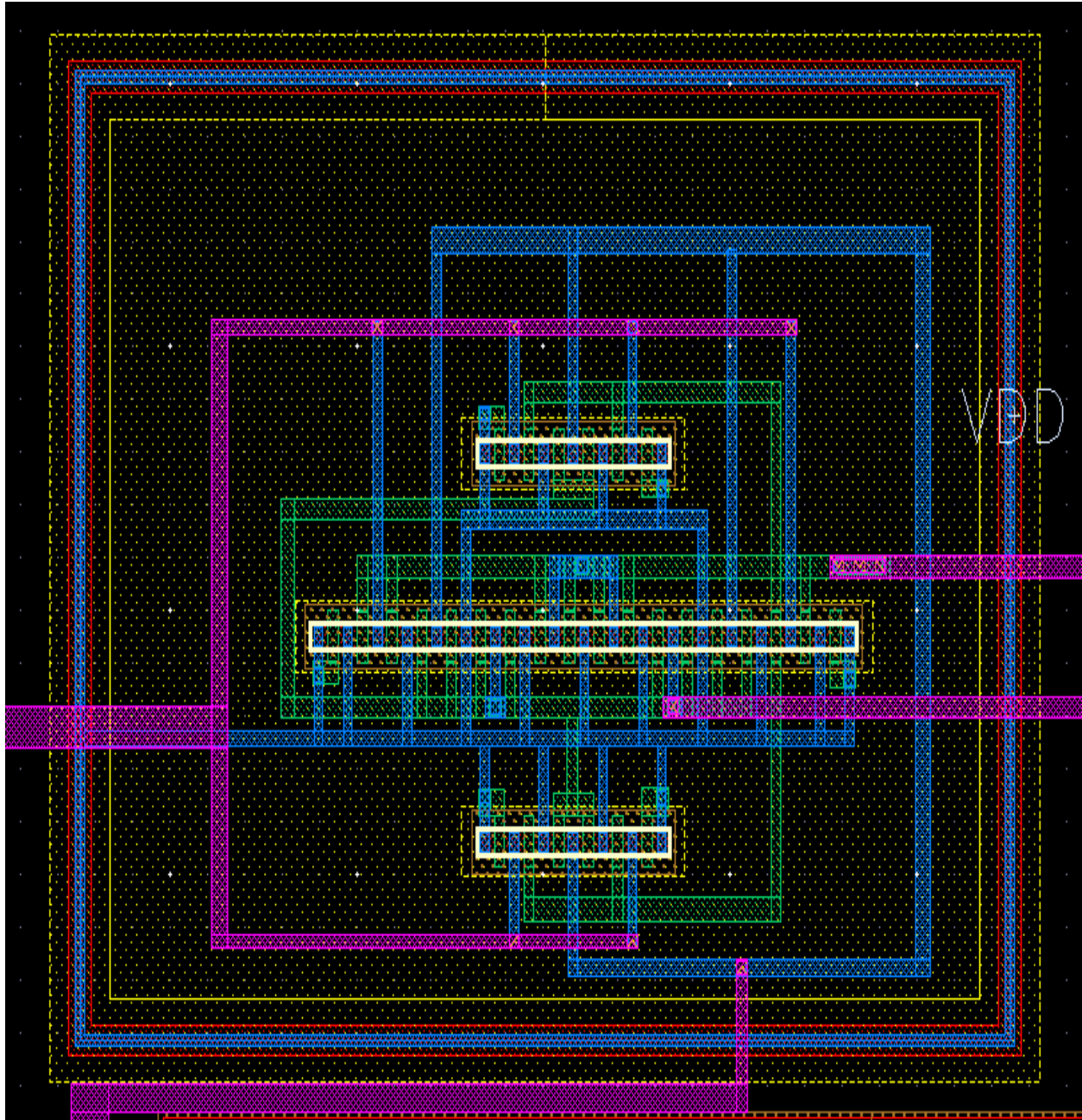
IV. LAYOUT(S) AND FLOOR PLAN(S)

Include diagrams of the floor plans used to lay out major circuit components. Typically, this will just include transistor floor plans (routing does not need to be included in the diagrams). A table can be used to show the layout arrangement for the transistors or the floor plan can be drawn in another software tool. Clearly label the devices in the floor plan and make sure there is clear correspondence with the devices in the schematics, and include dummy devices in the floor plan diagram (dummy devices are often denoted with an asterisk, *).

This section will also include images of the layout(s) of each circuit constructed for the lab.

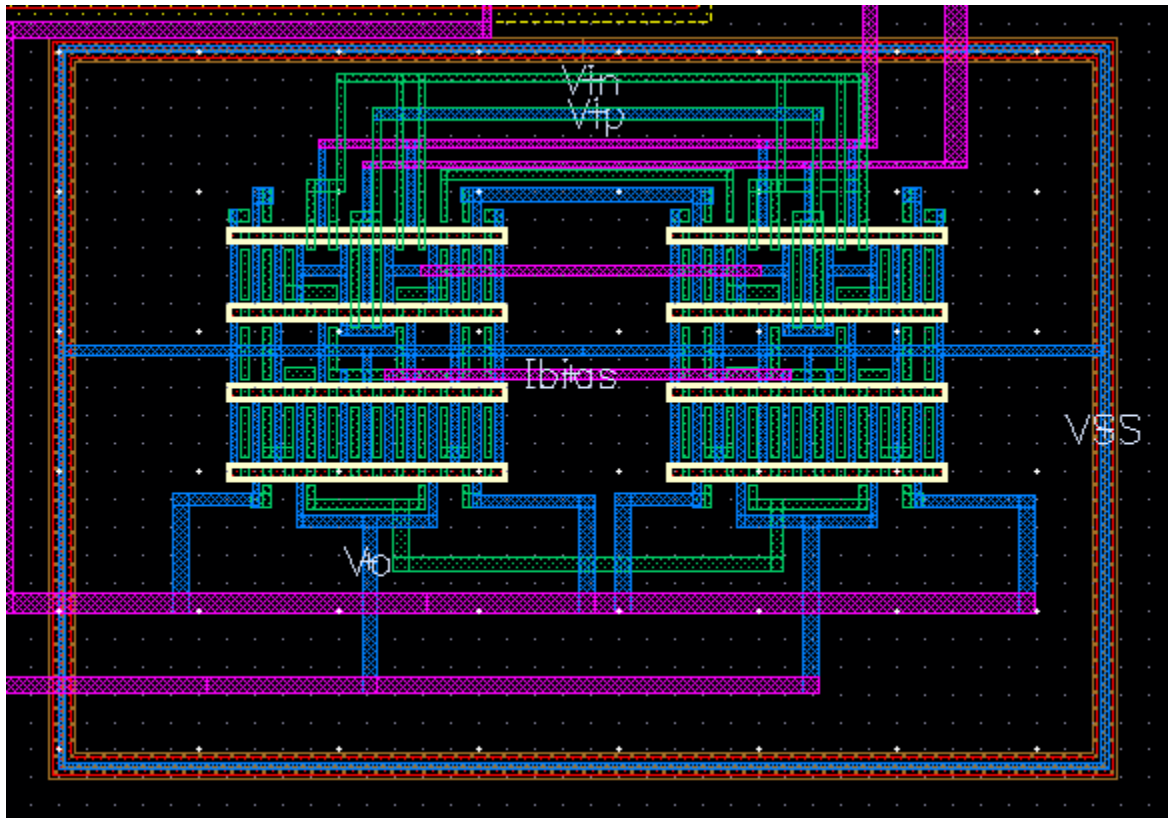
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CLOSER LOOK OF PMOS:

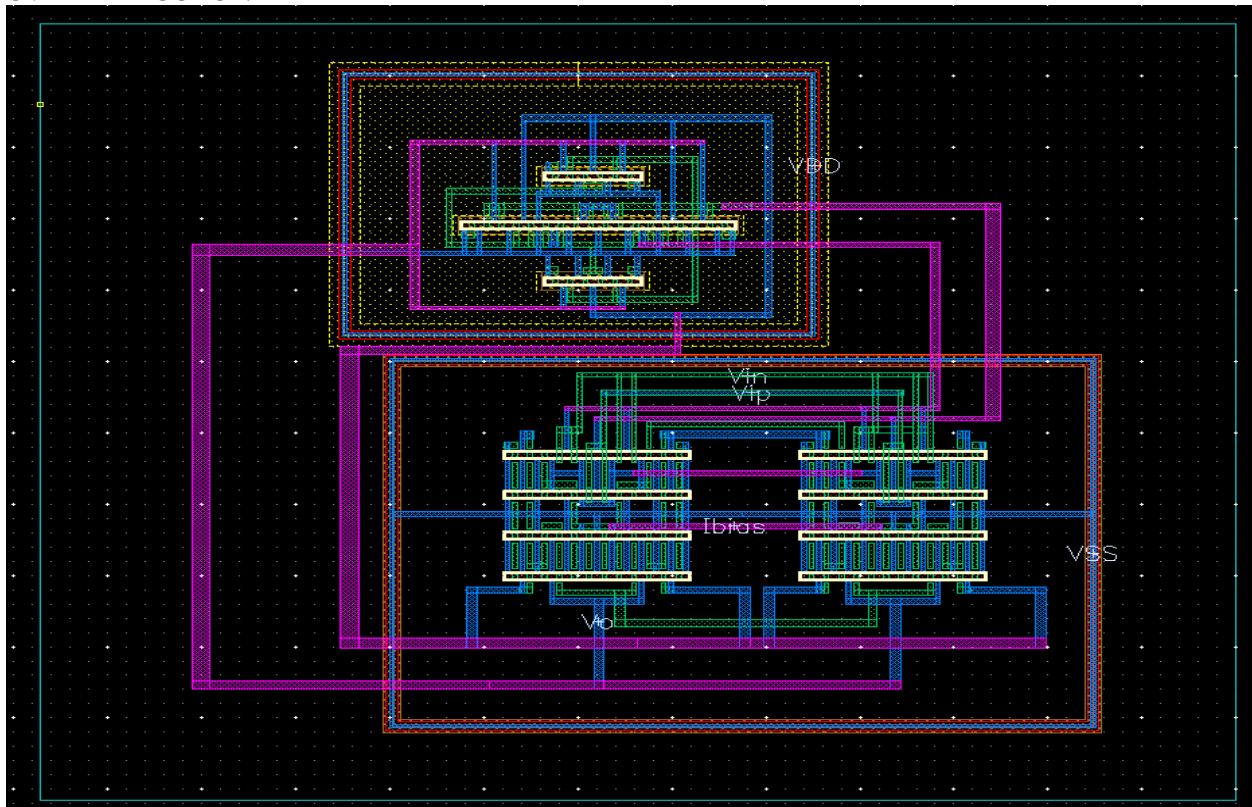


CLOSER LOOK OF NMOS:

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OVERALL LOOKUP:



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Floor plan

PMOS LAYOUT

M6M5M5M6

M6M6M5M5M3M3M4M4M4M4M3M3M5M5M6M6

M6M5M5M6

FLOOR PLAN:

NMOS LAYOUT

2 SEGMENTS OF SAME FLOORPLAN

M7M10M1M1M2M2M1M1M10M7

M7M10M1M1M2M2M1M1M10M7

M7M8M8M9M9M9M9M8M8M7

M7M8M8M9M9M9M9M8M8M7

M7M8M8M9M9M9M9M8M8M7

M7M8M8M9M9M9M9M8M8M7

M7M10M1M1M2M2M1M1M10M7

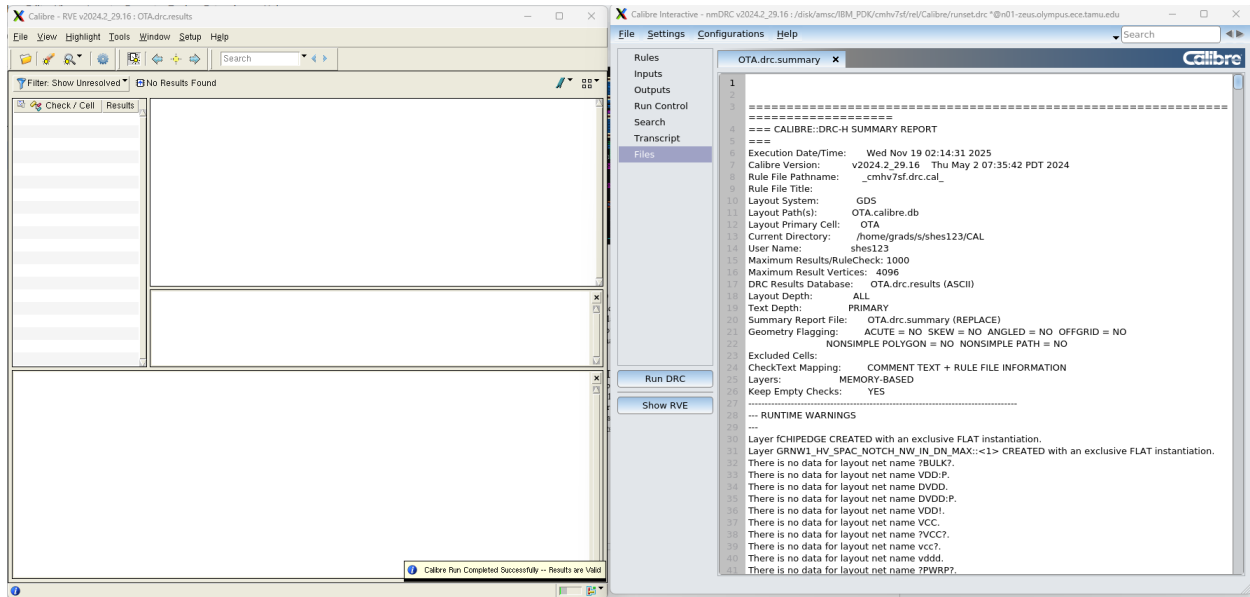
M7M10M1M1M2M2M1M1M10M7

V. DRC AND LVS RESULTS

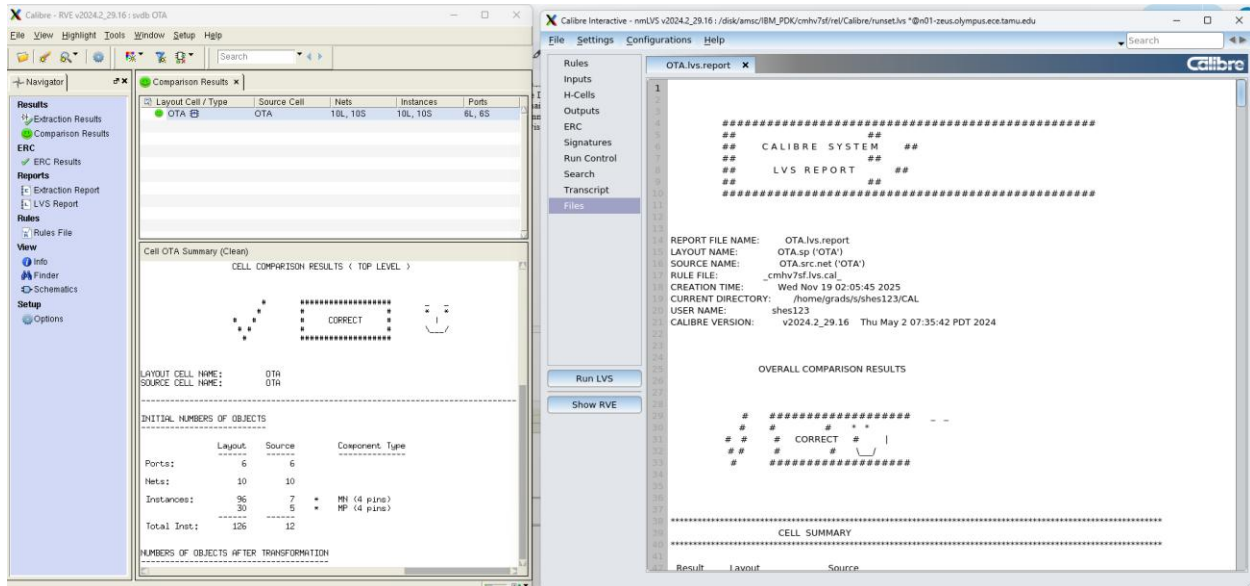
This section should include the DRC RVE window results for each layout cell (with the results filtered to “show unresolved” to show only the remaining DRC violations), and the cell name should be visible at the top of the DRC RVE window. The LVS report summary image should also be included for each cell with your NetID, the time stamp, and the smiley face/check mark visible to show that LVS passed.

DRC:

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LVS:



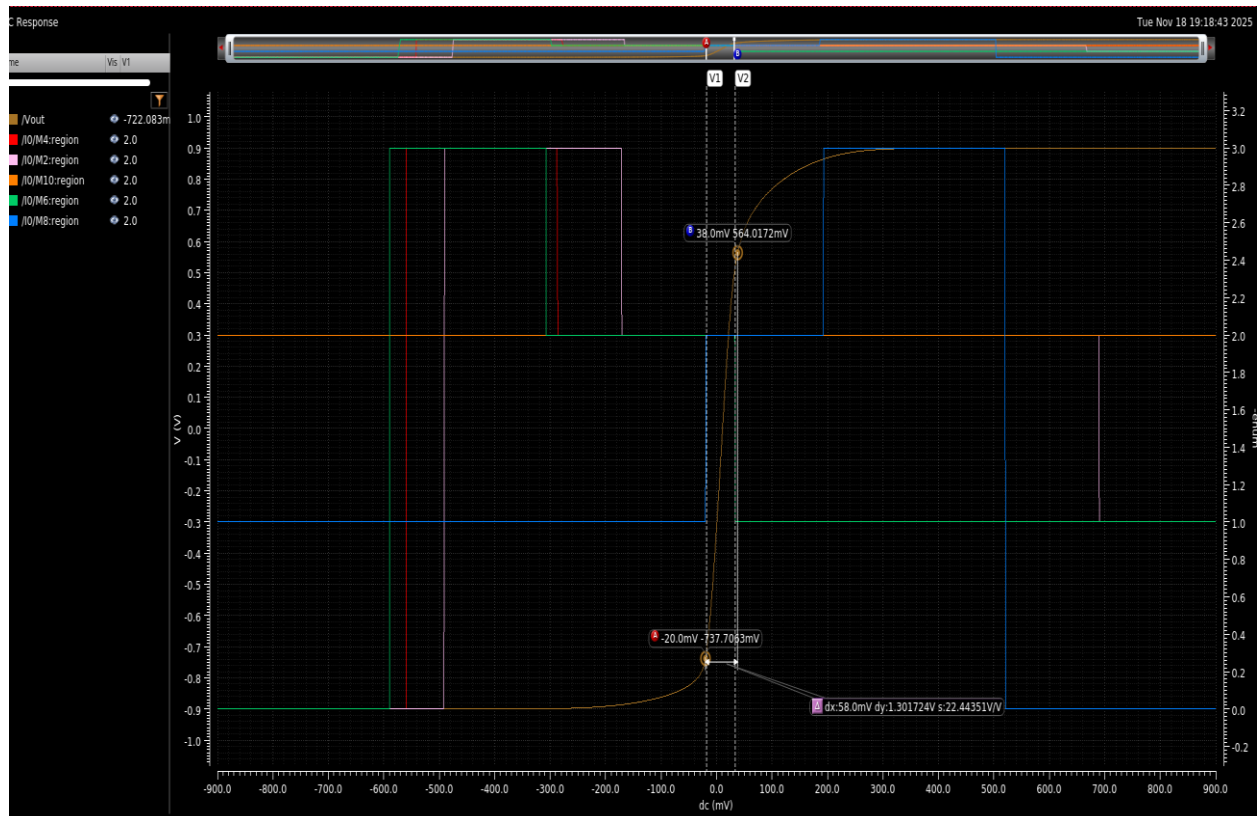
VI. SIMULATION RESULTS

Images of simulation results requested in the lab manual should be included here. Plots should be annotated to show relevant data points which will include either (1) data points that are explicitly requested in the lab manual or (2) data points from which other requested parameters are calculated. Data in the images should also be legible, so scale the images accordingly to make sure the data is easy to read. Post-layout simulations should be plotted in the same window with the pre-layout simulation results in order to make the differences between the two more apparent.

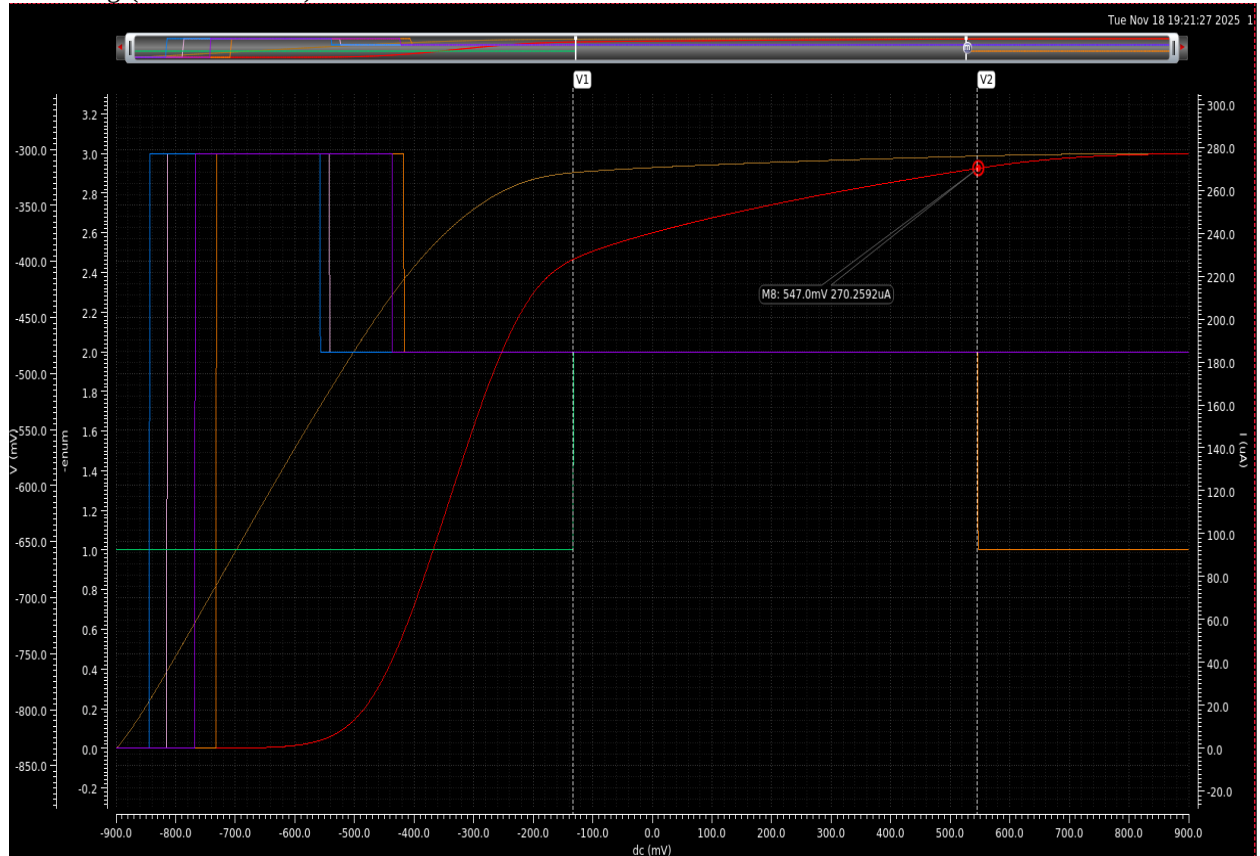
PRE-LAYOUT SIMULATION:

DC Swing:(common mode)

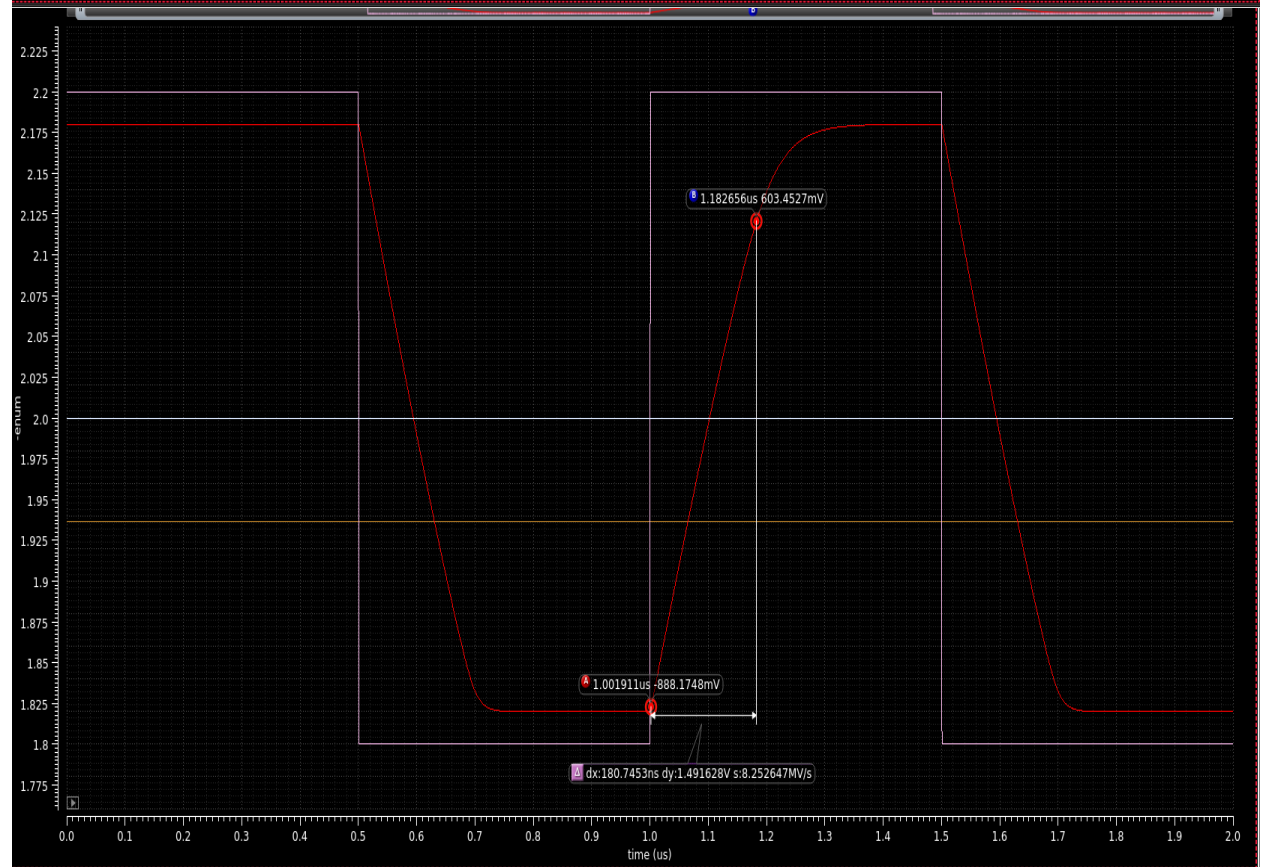
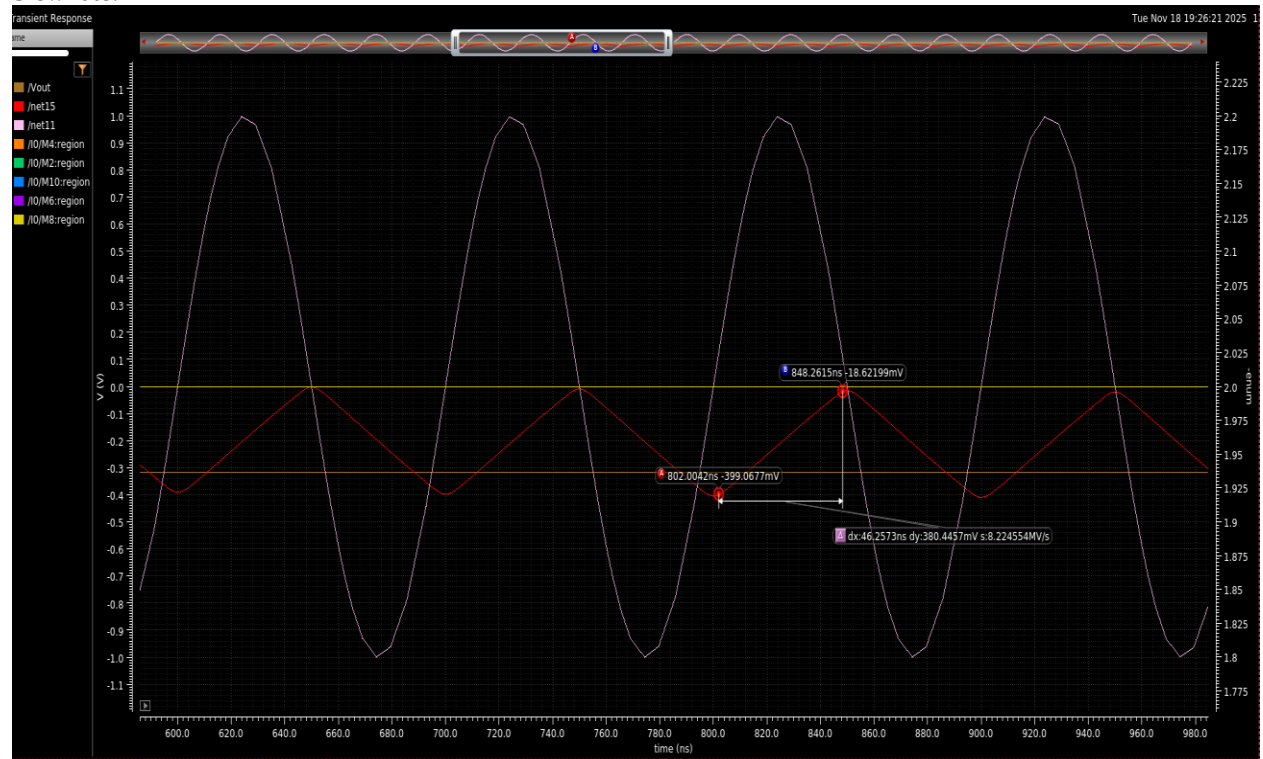
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DC Swing-(Common mode):

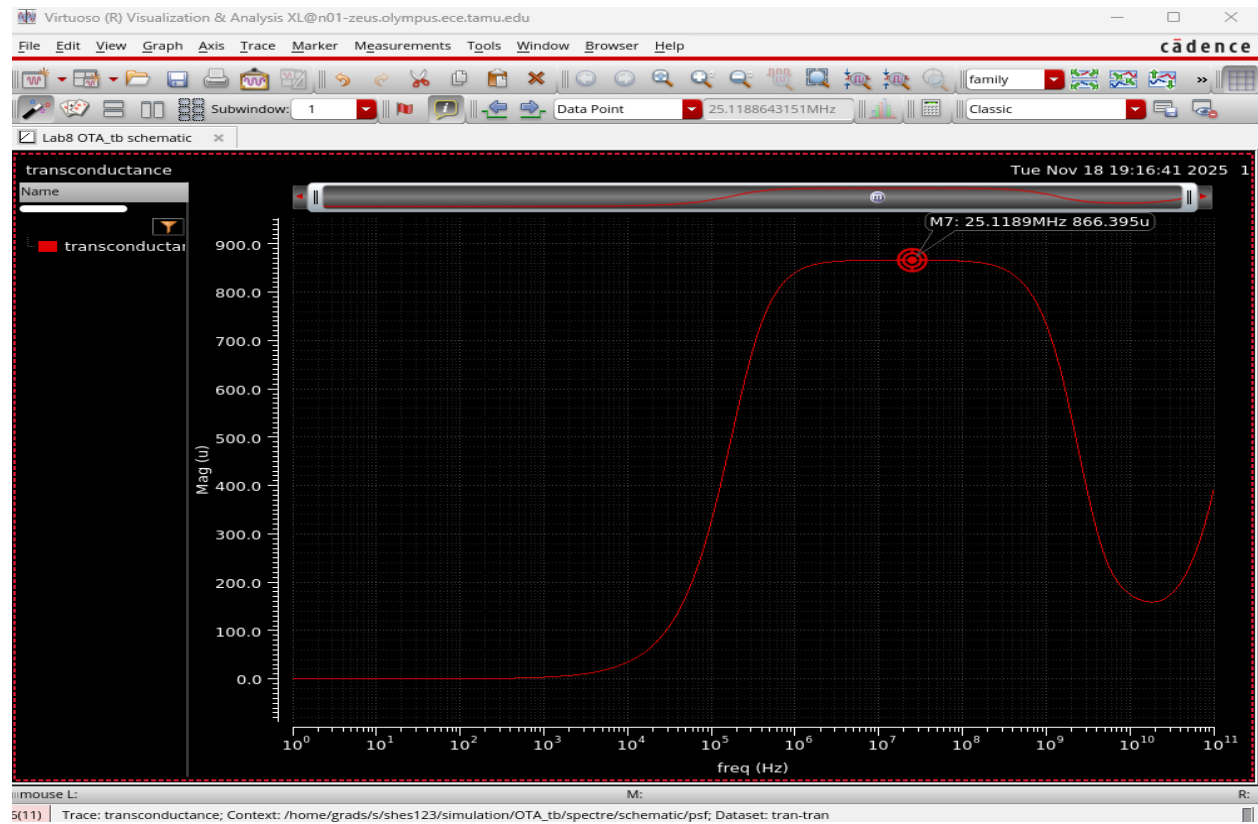


Slew rate:

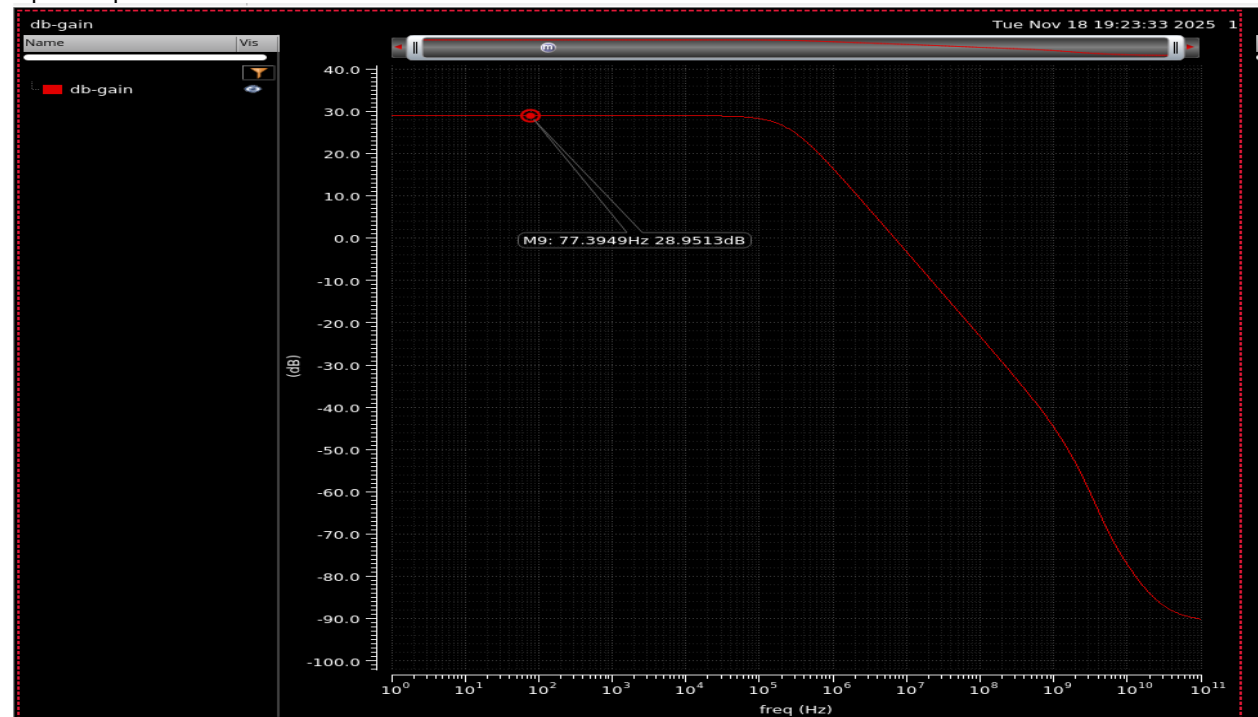


GM:

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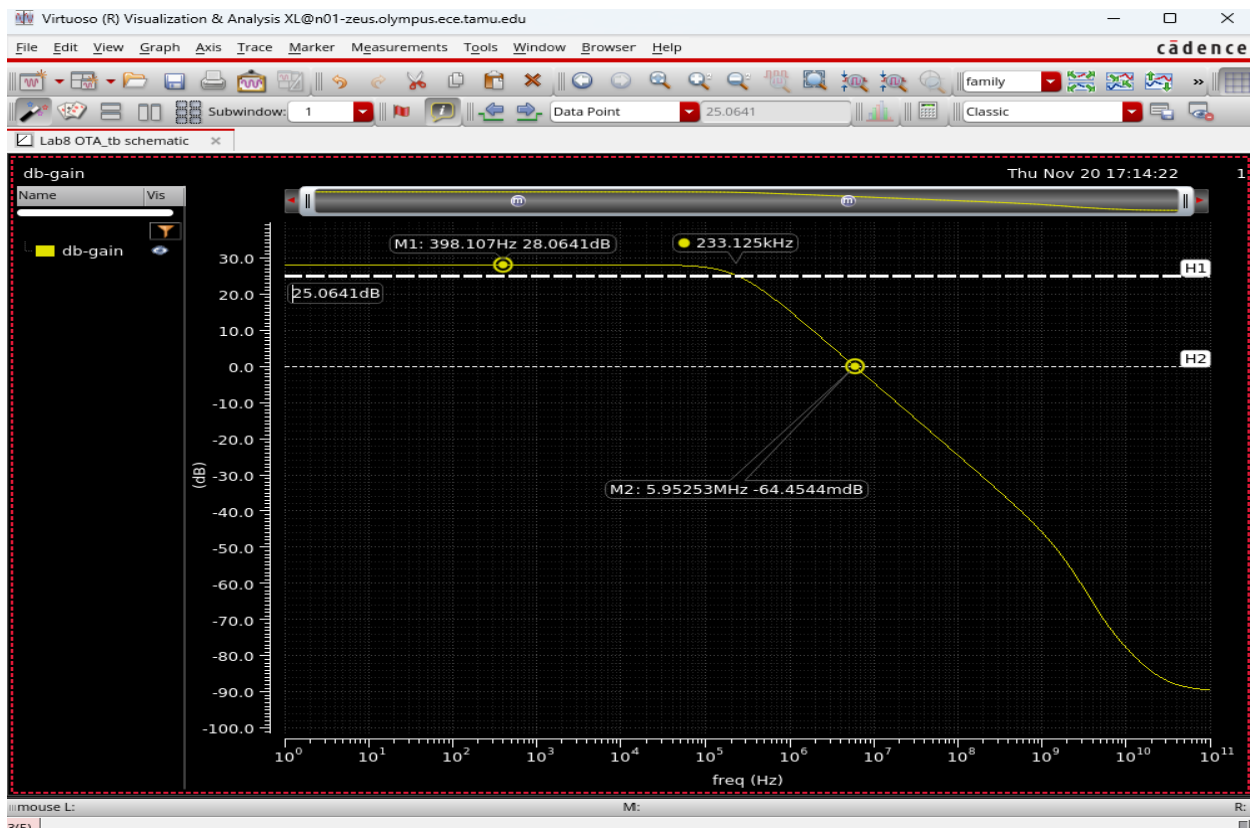
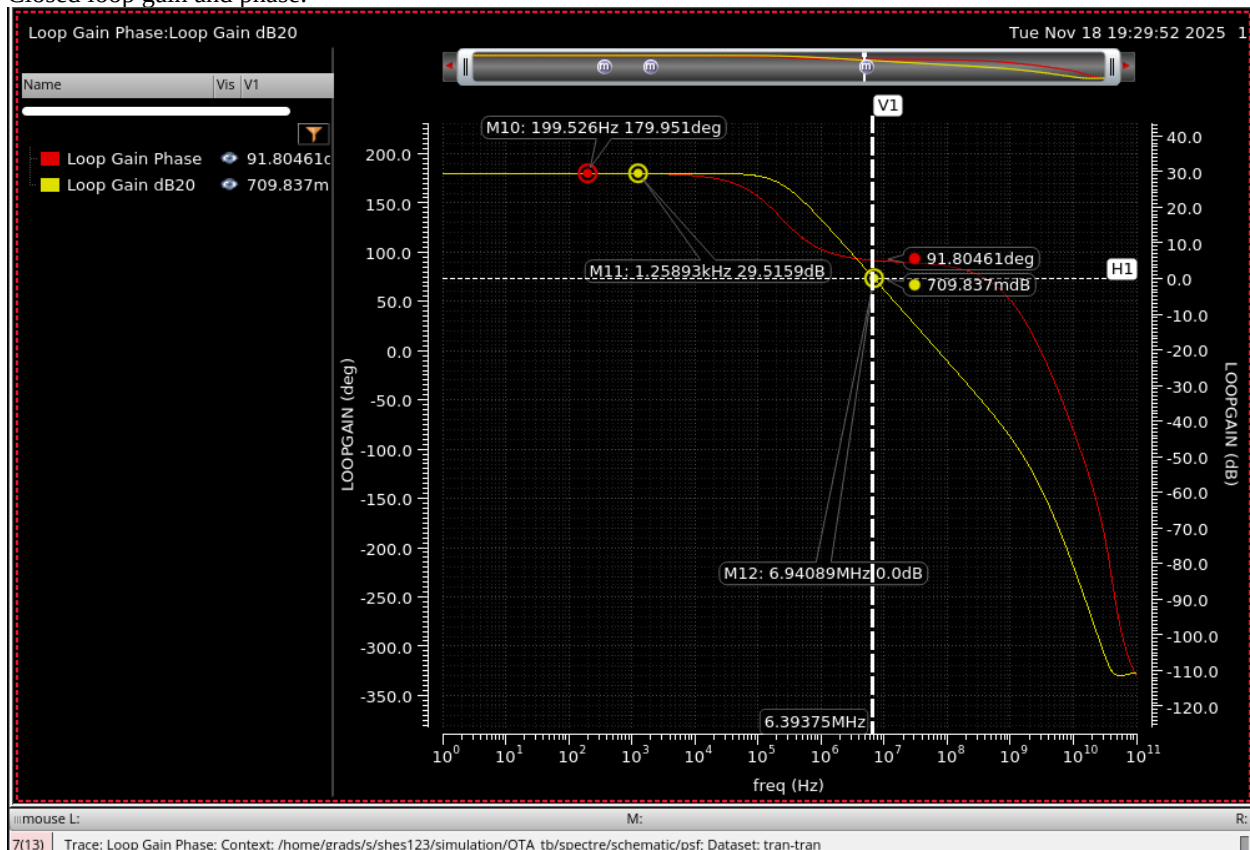
Open-loop Gain:



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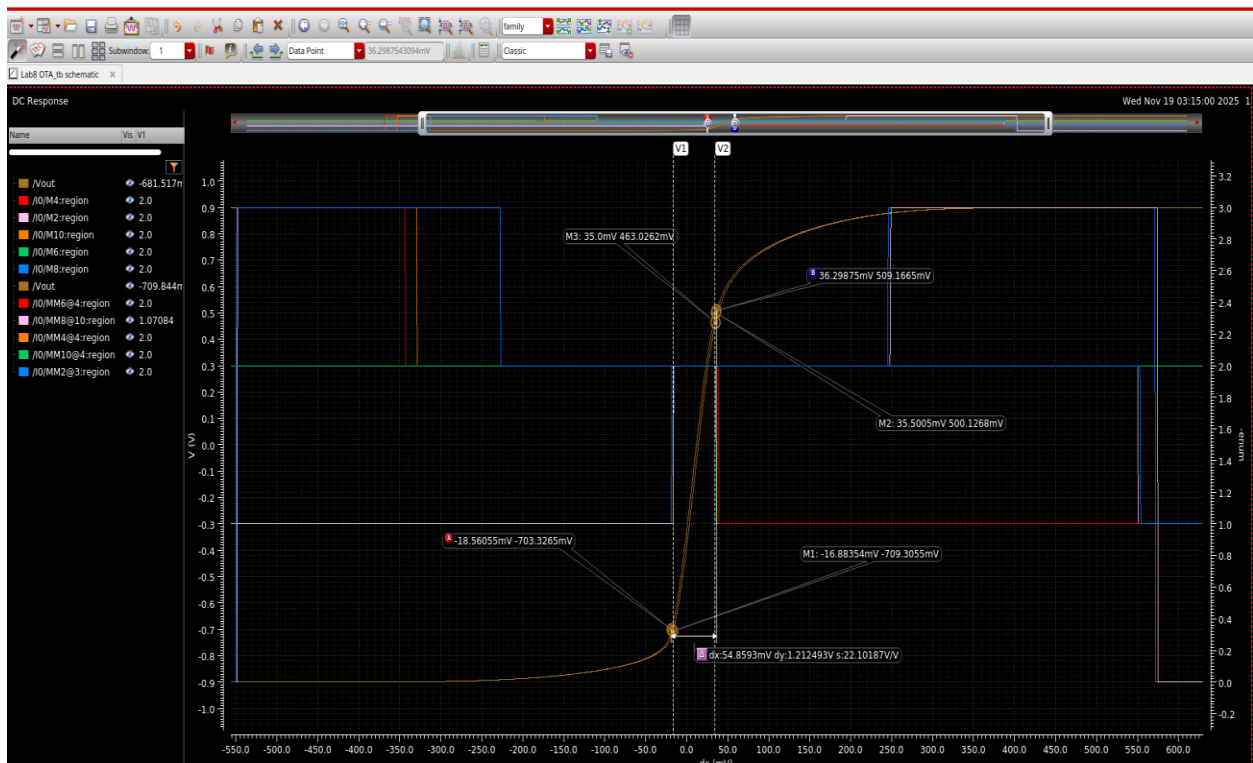
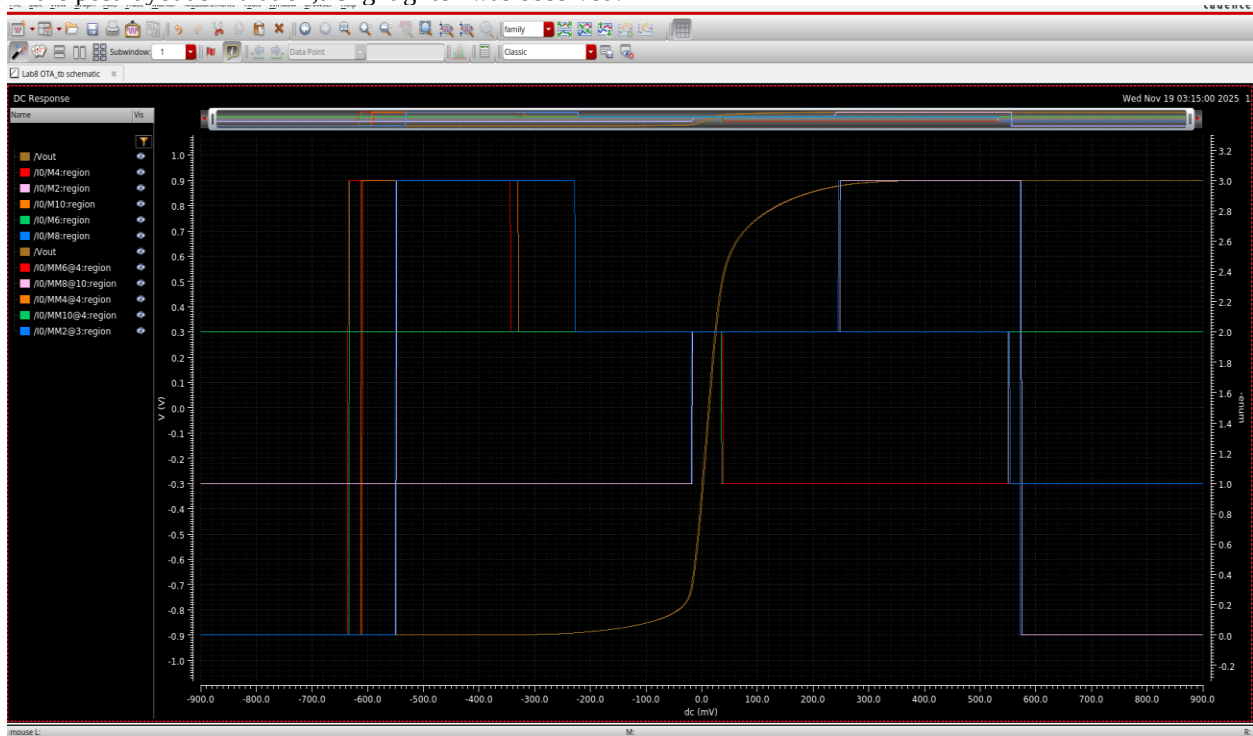
Closed loop gain and phase:



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POST-LAYOUT SIMULATION:
DC-DIFFERENTIAL MODE:

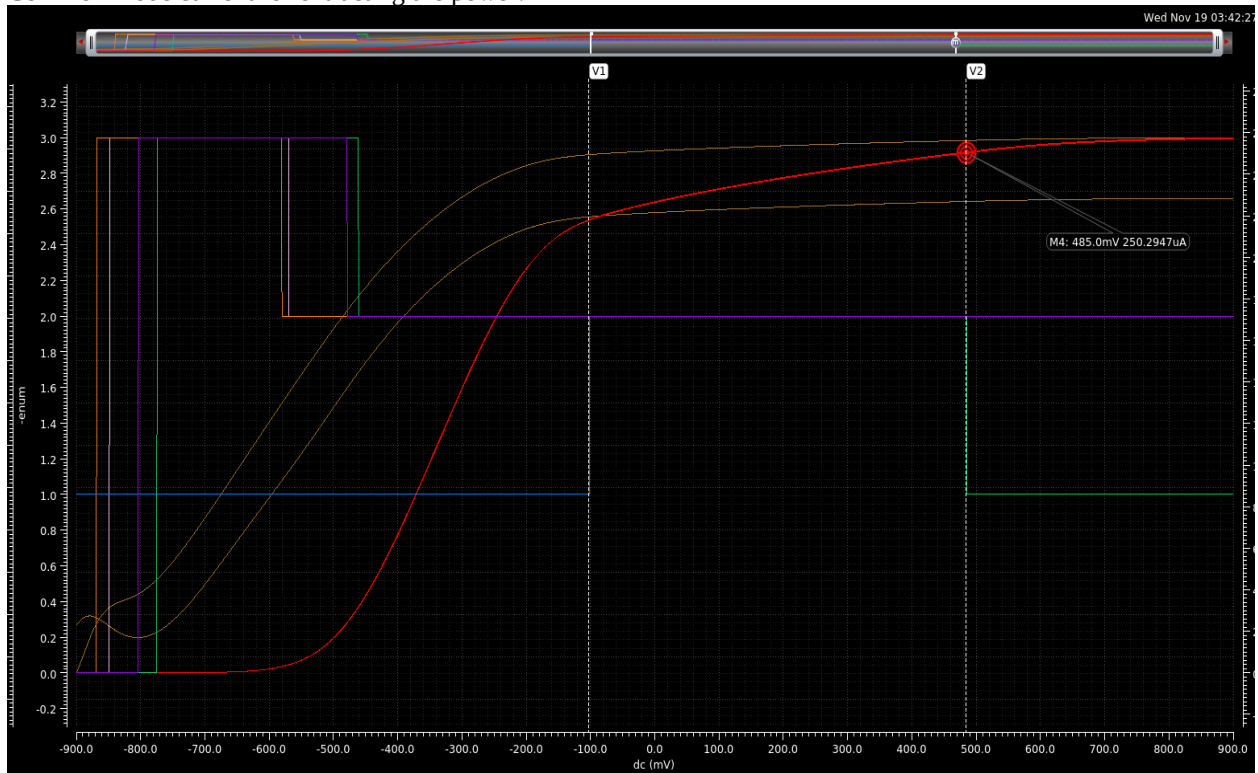
In the post layout simulation, a slight glitch was observed:



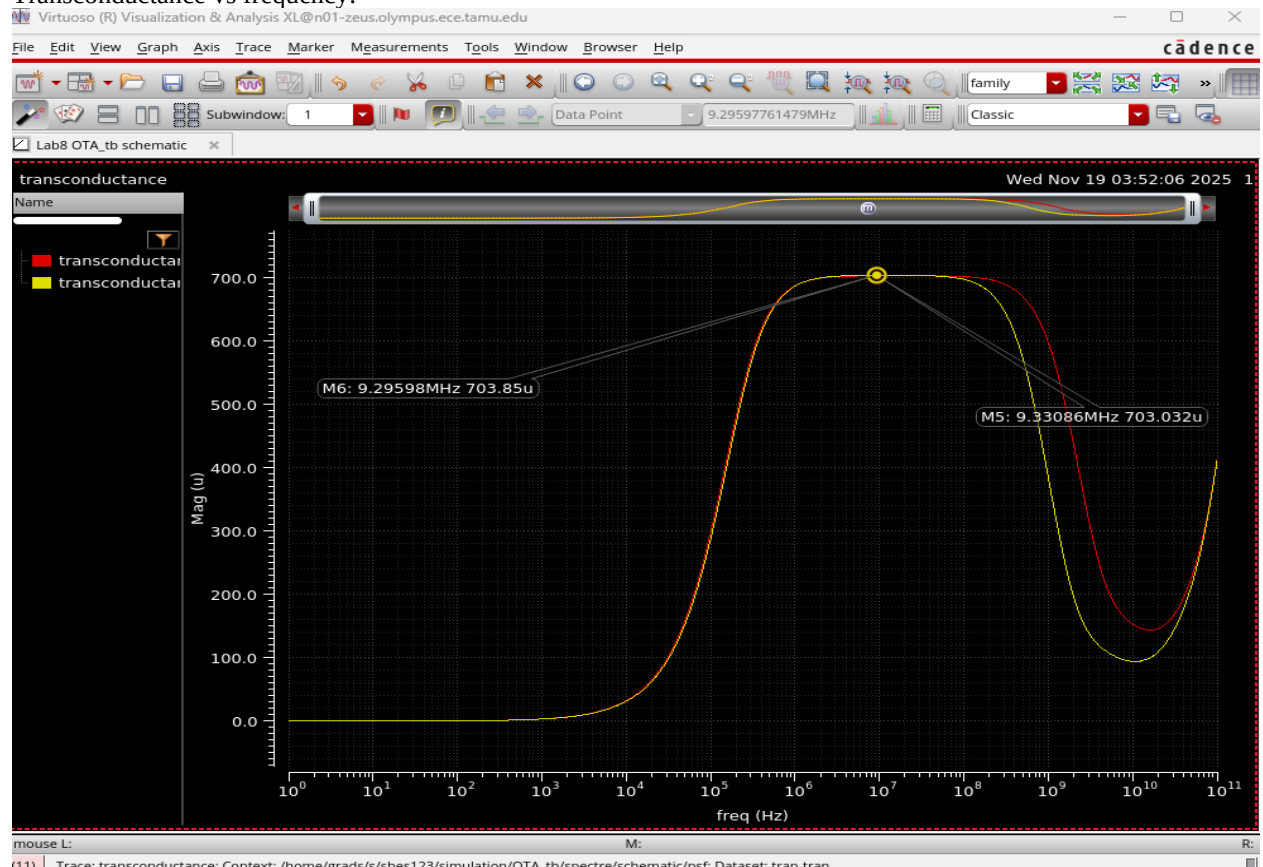
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Common mode current for extracting the power:

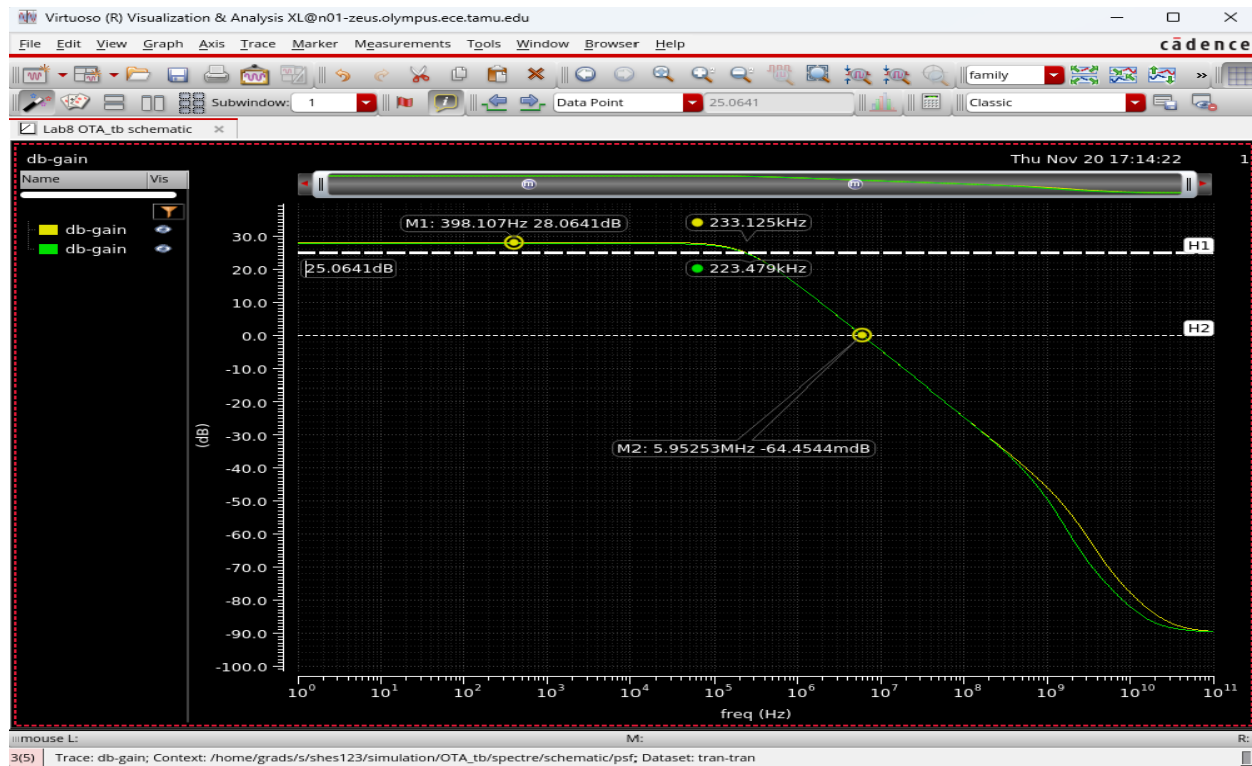
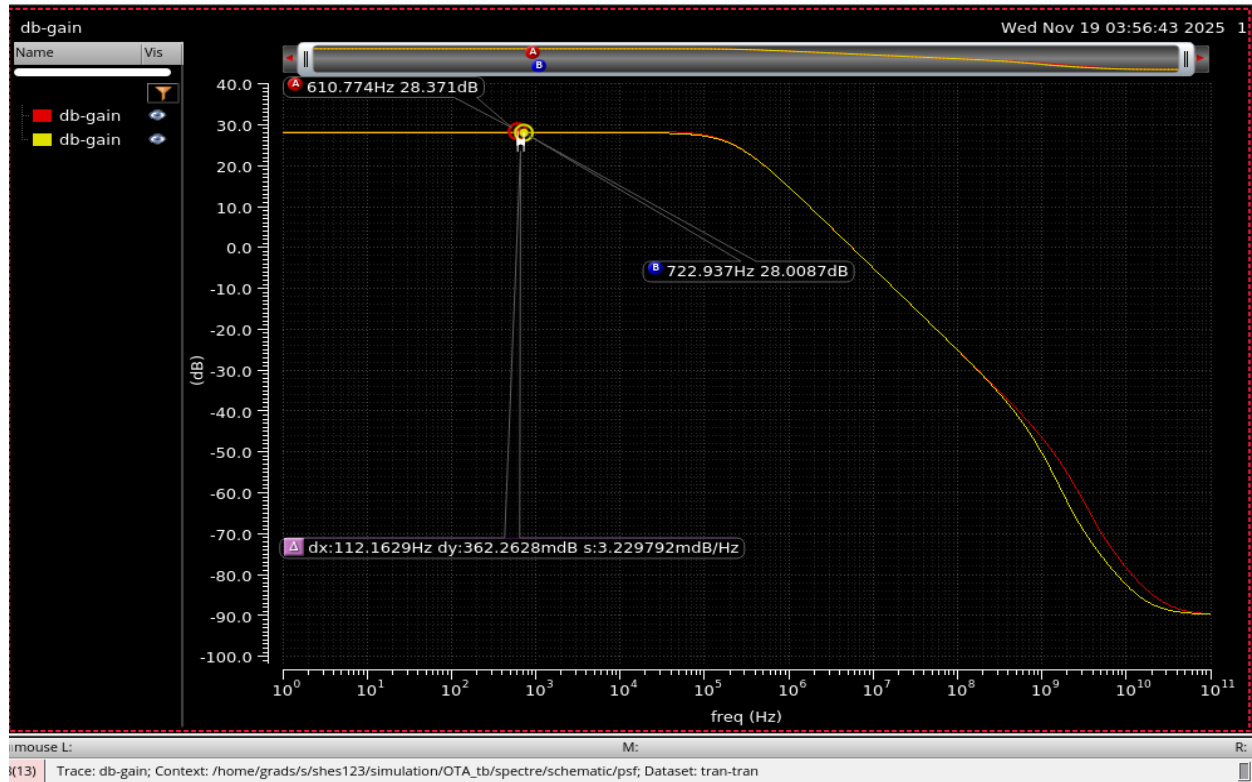


Transconductance vs frequency:

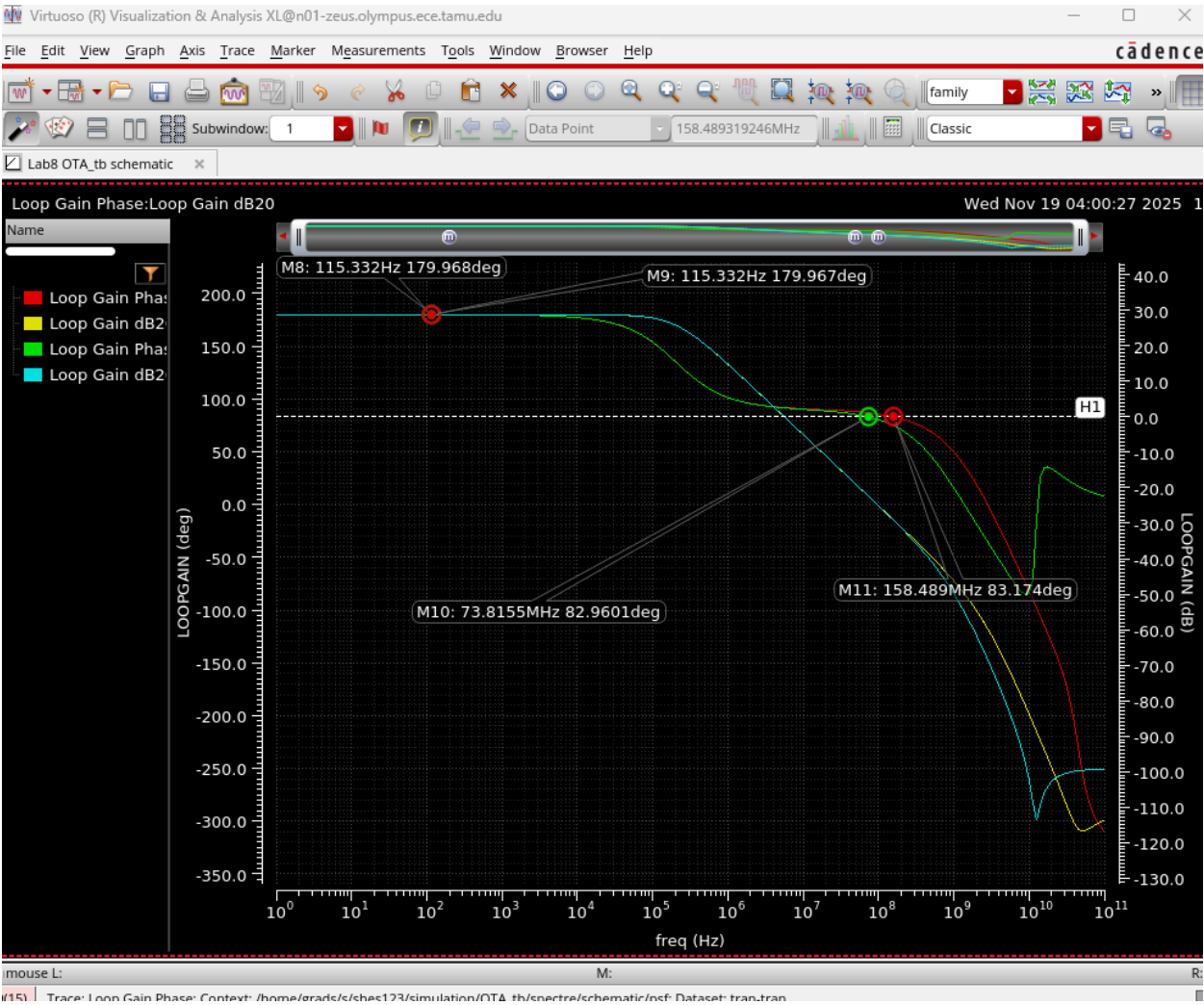


Open loop out put voltage /input voltage vs gain:

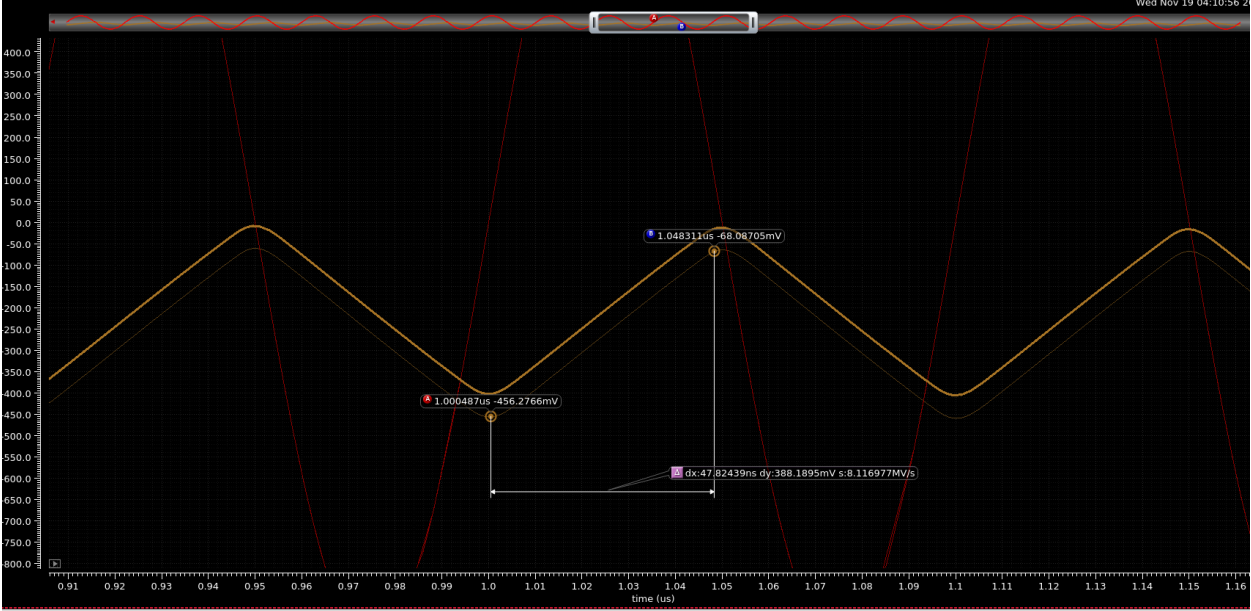
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Slew rate:



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